A Technical Introduction to Mean squared error

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Section 1

Definition and basic properties The MSE either assesses the quality of a predictor (i.e., a function mapping arbitrary inputs to a sample of values of some random variable), or of an estimator (i.e., a mathematical function mapping a sample of data to an estimate of a parameter of the population from which the data is sampled). In the context of prediction, understanding the prediction interval can also be useful as it provides a range within which a future observation will fall, with a certain probability. The definition of an MSE differs according to whether one is describing a predictor or an estimator.

Section 2

MSE ■ (θ ^) = E θ ■ [(θ ^ − θ) 2] = E θ ■ [(θ ^ − E θ ■ [θ ^] + E θ ■ [θ ^] − θ) 2] = E θ ■ [(θ ^ − E θ ■ [θ ^]) 2 + 2 (θ ^ − E θ ■ [θ ^]) (E θ ■ [θ ^] − θ) + (E θ ■ [θ ^] − θ) 2] = E θ ■ [(θ ^ − E θ ■ [θ ^]) 2] + E θ ■ [2 (θ ^ − E θ ■ [θ ^]) (E θ ■ [θ ^] − θ)] + E θ ■ [(E θ ■ [θ ^] − θ) 2] = E θ ■ [(θ ^ − E θ ■ [θ ^]) 2] + 2 (E θ ■ [θ ^] − θ) E θ ■ [θ ^ − E θ ■ [θ ^]] + (E θ ■ [θ ^] − θ) 2 E θ ■ [θ ^] − θ = constant = E θ ■ [(θ ^ − E θ ■ [θ ^]) 2] + 2 (E θ ■ [θ ^] − θ) (E θ ■ [θ ^] − E θ ■ [θ ^]) + (E θ ■ [θ ^] − θ) 2 E θ ■ [θ ^] = constant = E θ ■ [(θ ^ − E θ ■ [θ ^]) 2] + (E θ ■ [θ ^] − θ) 2 = Var θ ■ (θ ^) + Bias θ ■ (θ ^ , θ) 2 {\displaystyle {\begin{aligned}\operatorname {MSE} ({\hat {\theta }})&=\operatorname {E} _{\theta }\left[({\hat {\theta }}-\theta)^{2}\right]\\&=\operatorname {E} _{\theta }\left[\left({\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]+\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)^{2}\right]\\&=\operatorname {E} _{\theta }\left[\left({\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]\right)^{2}+2\left({\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]\right)\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)+\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)^{2}\right]\\&=\operatorname {E} _{\theta }\left[\left({\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]\right)^{2}\right]+\operatorname {E} _{\theta }\left[2\left({\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]\right)\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)\right]+\operatorname {E} _{\theta }\left[\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)^{2}\right]\\&=\operatorname {E} _{\theta }\left[\left({\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]\right)^{2}\right]+2\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)\operatorname {E} _{\theta }\left[{\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]\right]+\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)^{2}&&\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta ={\text{constant}}\\&=\operatorname {E} _{\theta }\left[\left({\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]\right)^{2}\right]+2\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\operatorname {E} _{\theta }[{\hat {\theta }}]\right)+\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)^{2}&&\operatorname {E} _{\theta }[{\hat {\theta }}]={\text{constant}}\\&=\operatorname {E} _{\theta }\left[\left({\hat {\theta }}-\operatorname {E} _{\theta }[{\hat {\theta }}]\right)^{2}\right]+\left(\operatorname {E} _{\theta }[{\hat {\theta }}]-\theta \right)^{2}\\&=\operatorname {Var} _{\theta }({\hat {\theta }})+\operatorname {Bias} _{\theta }({\hat {\theta }},\theta)^{2}\end{aligned}}}

Section 3

MSE ■ (θ ^) = E [(θ ^ − θ) 2] = Var ■ (θ ^ − θ) + (E [θ ^ − θ]) 2 = Var ■ (θ ^) + Bias 2 ■ (θ ^ , θ) {\displaystyle {\begin{aligned}\operatorname {MSE} ({\hat {\theta }})&=\mathbb {E} [({\hat {\theta }}-\theta)^{2}]\\&=\operatorname {Var} ({\hat {\theta }}-\theta)+(\mathbb {E} [{\hat {\theta }}-\theta])^{2}\\&=\operatorname {Var} ({\hat {\theta }})+\operatorname {Bias} ^{2}({\hat {\theta }},\theta)\end{aligned}}}

Section 4

Interpretation An MSE of zero, meaning that the estimator θ ^ {\displaystyle {\hat {\theta }}} predicts observations of the parameter θ {\displaystyle \theta } with perfect accuracy, is ideal (but typically not possible). Values of MSE may be used for comparative purposes. Two or more statistical models may be compared using their MSEs—as a measure of how well they explain a given set of observations: An unbiased estimator (estimated from a statistical model) with the smallest variance among all unbiased estimators is the best unbiased estimator or MVUE (Minimum-Variance Unbiased Estimator). Both analysis of variance and linear regression techniques estimate the MSE as part of the analysis and use the estimated MSE to determine the statistical significance of the factors or predictors under study. The goal of experimental design is to construct experiments in such a way that when the observations are analyzed, the MSE is close to zero relative to the magnitude of at least one of the estimated treatment effects. In one-way analysis of variance, MSE can be calculated by the division of the sum of squared errors and the degree of freedom. Also, the f-value is the ratio of the mean squared treatment and the MSE. MSE is also used in several stepwise regression techniques as part of the determination as to how many predictors from a candidate set to include in a model for a given set of observations.

Section 5

An even shorter proof can be achieved using the well-known formula that for a random variable X {\textstyle X} , E (X 2) = Var ■ (X) + (E (X)) 2 {\textstyle \mathbb {E} (X^{2})=\operatorname {Var} (X)+(\mathbb {E} (X))^{2}} . By substituting X {\textstyle X} with, θ ^ – θ {\textstyle {\hat {\theta }}-\theta } , we have